Asterix category 015 - Independent Non-Cooperative Surveillance System Target Reports

category: 015

edition: 1.2

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Preamble

Surveillance data exchange.

Description of standard data items

I015/000 - Message Type

definition: This data item conveys the report type and whether the output is periodically updated or asynchronous depending upon external events.

Group

I015/000/MT - Message Type

Element
bit size: 7
Values:

- **1:** Measurement Plot
- **2:** Measurement Track
- **3:** Sensor Centric Plot
- **4:** Sensor Centric Track
- **5:** Track End Message

I015/000/RG - Report Generation

Element
bit size: 1
Values:

- **0:** Periodic Report
- **1:** Event Driven Report

Note 1: See Section 4.7 and ANNEX A for definitions of the Message Types.

Note 2: Values 6 to 127 are reserved for future use

Note 3: Periodic Report: A periodic report is one transmitted periodically with an independently configurable period.

Note 4: Event Driven Report: An Event Driven Report is one generated in response to the occurrence of an external event such as an RF echo off a target.

I015/010 - Data Source Identifier

definition: Identification of the sensor from which the data is received.

Group

I015/010/SAC - System Area Code

Element
bit size: 8
Raw Content

I015/010/SIC - System Identification Code

Element
bit size: 8
Raw Content

Note 1: The up-to-date list of SACs is published on the EUROCONTROL Web Site (http://www.eurocontrol.int/asterix).

Note 2: The SICs are allocated by the national authority responsible for the surveillance infrastructure.

Note 3: The SIC and SAC values shall be formatted as binary unsigned integers.

I015/015 - Service Identification

definition: Identification of the service provided to one or more users.

Element
bit size: 8
Raw Content

Note 1: The Service Identification is allocated by the system.

Note 2: The SID value shall be formatted as binary unsigned integers.

I015/020 - Target Report Descriptor

definition: Type and characteristics of the data as transmitted by a system.

Extended

I015/020/MOMU - Mono-Static Target Report or Multi-Static Target Report

Element
bit size: 2
Values:

- **0:** Mono-Static Sensor
- **1:** Multi-Static Sensor
- **2:** Other
- **3:** Unknown

I015/020/TTAX - Target Taxonomy

Element
bit size: 2
Values:

- **0:** Actual Target Report
- **1:** Reference Target
- **2:** Synthetic Target
- **3:** Simulated / Replayed Target

I015/020/SCD - Scanning Direction

Element
bit size: 2
Values:

- **0:** Unknown
- **1:** Forward
- **2:** Backward
- **3:** Static

Spare bits: 1

(FX) - extension bit

Note 1: The MoMu bit is used to indicate whether the target report was constructed from a multi-static (including bi-static) or mono-static sensor. Its setting dictates the interpretation of data items I015/625 and I015/626. The meaning of the value “other” shall be described in the system ICD.

Note 2: In this context, a Reference Target Report stems from a non-aircraft target based on RF received externally to the system boundary. This may be generated, for example, by an external RF generator or a Permanent Echo or from a device, which is deployed in line of sight of the sensor.

Note 3: A synthetic target is an internally generated diagnostic signal prior to the generation of the ASTERIX Category 015 target report. For example used to support test processes.

Note 4: This value is used to represent externally generated targets or recorded data injected into the output data stream of the INCS system e.g. for test or training purposes.

Note 5: This indication is used to inform about the scanning direction of the system (e.g. left/right, up/down, clockwise/anti-clockwise). It’s exact meaning is implementation dependent and shall be described in the system ICD.

I015/030 - Warning/Error Conditions

definition: Warning/error conditions detected by a system for the target report involved.

Repetitive

With FX extension bit.

Element
bit size: 7
Raw Content

Note 1: It has to be stressed that a series of one or more W/E conditions can be reported per target report.

Note 2: The nature of the warning / error condition may differ between sensor types and the declaration and use of such alerts is driven by end user requirements.

Note 3: Potential applications could be to indicate that the target report correlates with road infrastructure (terrestrial vehicles) or a wind turbine or that it is a fixed or slow moving return or originating from an area of high clutter. Such data items could also be used to indicate the presence of interference – either deliberate or accidental.

Note 4: The Warning/Error Condition Values from 1-31 are reserved for designation by the ASTERIX Maintenance Group. System implementers are free to use values of 32 and above. The allocation of the remaining values of this data item shall be defined in a local Interface Control Document.

Note 5: The value of “0” must not be assigned.

I015/050 - Update Period

definition: Period until next expected output of a target report for this target.

Group

Spare bits: 2

I015/050/UPD - Update Period

Element
bit size: 14
Unsigned quantity
LSB = $1/2^7$ s ≈ $7.8125e-3$ s
unit: ”s”
<= 128.0

Note 1: It is not necessary that all targets detected by the INCS sensor have target reports generated at the same update period.

Note 2: This data item indicates the period until the next expected output of a target report for this target relative to the Time of Applicability contained in data item I015/145

I015/145 - Time of Applicability

definition: Absolute time stamping for applicability of the measured information expressed as UTC.

Element
bit size: 24
Unsigned quantity
LSB = $1/2^7$ s ≈ $7.8125e-3$ s
unit: ”s”
< 86400.0

Note 1: The Time of Applicability refers to the information contained in data item I015/600, I015/601, I015/625, I015/626, I015/627, I015/628 whichever is available. In case of a Track End Message (Message Type = 5) it refers to the time at which the track is terminated and the track number (data item I015/161) is released for re-use.

Note 2: A distributed sensor, such as an MSPSR, may have multiple elements that are each individually time stamped which are consolidated in to a target report. Rather than provide details of each time stamped message, this data item conveys the time of applicability of position of the target report.

Note 3: The Time of Applicability value is reset to zero each day at midnight.

Note 4: The Time of Applicability value shall be formatted as a binary unsigned integer.

I015/161 - Track/Plot Number

definition: An integer value representing a unique reference to a track/plot record.

Element
bit size: 16
Unsigned integer
<= 65535.0

Note 1: Track numbers are required for Sensor and Measurement Tracks. However, for Sensor and Measurement Plots the inclusion of a track number is optional – depending upon whether the INCS sensor has used tracking processing to reduce the false alarm rates.

Note 2: The track number is allocated by the system.

Note 3: The track number value shall be formatted as binary unsigned integers.

I015/170 - Track/Plot Status

definition: Status of Track/Plot.

Extended

I015/170/BIZ

Element
bit size: 1
Values:

0: Target not in Blind Zone
1: Target in Blind Zone

I015/170/BAZ

Element
bit size: 1
Values:

0: Target not in Blanked Zone
1: Target in Blanked Zone

I015/170/TUR

Element
bit size: 1
Values:

0: Track Alive
1: Track Terminated by User Request

Spare bits: 1

I015/170/CSTP - Coasted - Position

Element
bit size: 1
Values:

0: Not extrapolated
1: Extrapolated

I015/170/CSTH - Coasted – Height

Element
bit size: 1
Values:

0: Not extrapolated
1: Extrapolated

I015/170/CNF - Confirmed vs. Tentative Track

Element
bit size: 1
Values:

0: Confirmed Track
1: Tentative Track

(FX) - extension bit

Note 1: The indication for CSTP and/or CSTH applies only to data items I015/600 and I015/605 respectively. In case one of these data items is not present, CSTP and/or CSTH has no meaning.

Note 2: A coasted track is one for which the sensor detections have been interrupted and whose position/height is being predicted based on the previously received responses.

Note 3: The blind zone or blanked zone are predictable zones where no detection is predicted. If bit 5 is set and TTS = 1 then the track is coasted because it is in a blind zone or sector blank zone.

Note 4: The indication TUR=1 shall be sent only with Message Type = 5 “Track End Message”.

I015/270 - Target Size & Orientation

definition: Data item containing the size and orientation information of the target.

Compound

I015/270/LEN - Target Length

description: The target length is the longest dimension in the targets direction of motion

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m $\approx 1.0e-2$ m
unit: ”m”
<= 655.35

I015/270/WDT - Target Width

description: The target width is the longest dimension orthogonal to the targets direction of motion

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m $\approx 1.0e-2$ m
unit: ”m”
<= 655.35

I015/270/HGT - Target Height

description: The target height is the longest dimension in the vertical direction.

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m $\approx 1.0e-2$ m
unit: ”m”
<= 655.35

I015/270/ORT - Target Orientation

description: The orientation gives the direction, which the target nose is pointing, relative to the Geographical North.

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° $\approx 5.4931640625e-3$ °
unit: ”°”
<= 360.0

Note: The orientation shall increment in a clockwise manner relative to Geographic North.

Note: If length and width cannot be clearly determined, the greater value of the two shall be transmitted as length.

I015/300 - Object Classification

definition: Classification result of the object detection.

Repetitive

Regular, 1 byte(s) REP field size.

Group

I015/300/CLS - Classification

Element
bit size: 9
Unsigned integer

I015/300/PRB - Probability

Element
bit size: 7
Unsigned integer

Note 1: INCS processing may be able to provide an indication of the nature of the target e.g. road vehicle or aircraft with the potential for further discrimination in the type of the aircraft e.g. two engine, fixed wing/helicopter etc. Before including requirements for target classification it is necessary to consider the operational manner in which such information would be used and how/if such data would be made available to the controllers.

Note 2: Target classification is attributing, with an acceptable degree of confidence, a target report as having originated from a specific object or target type e.g. fixed wing aircraft, flock of birds etc. (It should be noted that the probabilities assigned to a target do not necessarily need to add up to 100%) The ability of an INCS sensor to classify the targets it detects is dependent upon the systems capabilities and is driven by end user requirements. The use of this optional data item is to be agreed between parties such as the system manufacturer, the system operator agency and the end user. The allocation of the CLS octet is to be defined in a local Interface Control Document that shall be agreed by both parties.

Note 3: If this functionality is implemented in the sensor, the classification result (CLS), PRB and REP value shall be formatted as a binary unsigned integer.

I015/400 - Measurement Identifier

definition: An identifier pointing to a measurement that was created from a specific contributing Tx/Rx Pair where the Pair Identifier refers to the index which details both the transmitter characteristics (DVB-T, DAB, FM, dedicated etc.) and the receiver characteristics. These are defined in ASTERIX Category 016 – Data Item I016/300).

Group

I015/400/PID - Pair Identifier

Element
bit size: 16
Unsigned integer

I015/400/ON - Observation Number

Element
bit size: 24
Unsigned integer

Note 1: INCS sensors may achieve their operational requirements based upon different techniques and technologies. Some may utilise multiple transmitter stations or multiple receiver stations. This data items provides the means for subsequent processing stages to be able to analyse the target report data based upon the system components that contributed to the formation of the target report. See ANNEX A for further details.

Note 2: The Pair Identifier shall be defined in ASTERIX Category 016 (Data Item I016/300).

Note 3: The Observation Number is identifying an element of ‘raw data’ information. The sensor plots and sensor tracks are usually composed of several

I015/480 - Associations

definition: Information on which Measurement Identifiers contributed to the Sensor Centric Plot / Sensor Centric Track.

Repetitive

Regular, 1 byte(s) REP field size.

Element
bit size: 40
Raw Content

I015/600 - Horizontal Position Information

definition: Data item containing the horizontal position information of the target.
Compound

I015/600/P84 - Horizontal Position in WGS-84 Coordinates

description: Position of a target in WGS-84 Coordinates.
Group

I015/600/P84/LATITUDE

description: Latitude in WGS-84, in Two's complement.
Element
bit size: 32
Signed quantity
LSB = $180/2^31$ ° ≈ $8.381903171539306640625e-8$ °
unit: "°"
>= -90.0
< 90.0

I015/600/P84/LONGITUDE

description: Longitude in WGS-84, in Two's complement.
Element
bit size: 32
Signed quantity
LSB = $180/2^31$ ° ≈ $8.381903171539306640625e-8$ °
unit: "°"
>= -180.0
< 180.0

Note: The LSB provides a resolution of 1cm. Positive longitude indicates East. Positive latitude indicates North.

I015/600/HPR - Horizontal Position Resolution

description: A horizontal 2D dimensional area (ellipse) within which the sensor is unable to resolve two separate targets.
Group

I015/600/HPR/RSHPX

description: Horizontal position resolution of the target in target centric Cartesian coordinates (X-component)
Element
bit size: 16
Unsigned quantity
LSB = $1/2$ m ≈ 0.5 m
unit: "m"
<= 32767.0

I015/600/HPR/RSHPY

description: Horizontal position resolution of the target in target centric Cartesian coordinates (Y-component)
Element
bit size: 16
Unsigned quantity
LSB = $1/2$ m ≈ 0.5 m
unit: "m"
<= 32767.0

I015/600/HPR/CORSHPXY

description: Correlation of horizontal position resolution of X and Y components, in Two's complement.
Element
bit size: 8
Signed quantity
LSB = $1/2^7$ ≈ $7.8125e-3$
unit: ""

I015/600/HPP - Horizontal Position Precision

description: The distribution of horizontal position random errors.
Group

I015/600/HPP/SDHPX

description: Standard Deviation of horizontal position of the target in target centric Cartesian coordinates (X-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/2^2$ m ≈ 0.25 m
unit: ”m”
<= 16383.75

I015/600/HPP/SDHPY

description: Standard Deviation of horizontal position of the target in target centric Cartesian coordinates (Y-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/2^2$ m ≈ 0.25 m
unit: ”m”
<= 16383.75

I015/600/HPP/COSDHPXY

description: Correlation of standard deviation of horizontal position of X and Y components, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/601 - Geometric Height Information

definition: Data item containing the geometric height information of the target in WGS 84 height above ellipsoid.

Compound

I015/601/GH - Geometric Height (WGS-84)

description: Vertical distance between the target and the projection of its position on the earth’s ellipsoid, as defined by WGS-84, in Two’s complement form.

Element
bit size: 24
Signed quantity
LSB = $1/100$ m ≈ $1.0e-2$ m
unit: ”m”
>= -83286.0
<= 83286.0

I015/601/RSGH - Geometric Height Resolution

description: Vertical distance within which the sensor is unable to resolve two separate targets.

Element
bit size: 24
Unsigned quantity
LSB = $1/100$ m ≈ $1.0e-2$ m
unit: ”m”
<= 167772.15

I015/601/SDGH - Geometric Height Precision

description: The distribution of random Geometric Height errors (see also the definition of Precision in the appendix).

Element
bit size: 24
Unsigned quantity
LSB = $1/100$ m ≈ $1.0e-2$ m
unit: ”m”
<= 167772.15

I015/601/CI6 - Confidence Interval for Geometric Height (67%)

description: A measure of the uncertainty within which 67% of geometric height measurements will be contained.
Group

I015/601/CI6/UCI6

description: Upper confidence interval for Geometric Height (67%)

Element
bit size: 12
Unsigned quantity
LSB = 16 m ≈ 16.0 m
unit: ”m”
<= 65520.0

I015/601/CI6/LCI6

description: Lower confidence interval for Geometric Height (67%)

Element
bit size: 12
Unsigned quantity
LSB = 16 m ≈ 16.0 m
unit: ”m”
<= 65520.0

I015/601/CI9 - Confidence Interval for Geometric Height (95%)

description: A measure of the certainty within which 95% of geometric height measurements will be contained.
Group

I015/601/CI9/UCI9

description: Upper confidence interval for Geometric Height (95%)

Element
bit size: 12
Unsigned quantity
LSB = 16 m ≈ 16.0 m
unit: ”m”
<= 65520.0

I015/601/CI9/LCI9

description: Lower confidence interval for Geometric Height (95%)

Element
bit size: 12
Unsigned quantity
LSB = 16 m ≈ 16.0 m
unit: ”m”
<= 65520.0

I015/601/COGHHP - Correlation of Geometric Height and Horizontal Position

description: Correlation of Geometric Height converted into metres and Horizontal Position of X/Y-components.
Group

I015/601/COGHHP/X

description: Correlation of Geometric Height converted into metres and Horizontal Position of (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/601/COGHHP/Y

description: Correlation of Geometric Height converted into metres and Horizontal Position of (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/601/COGHHV - Correlation of Geometric Height and Horizontal Velocity

description: Correlation of Geometric Height converted into metres and Horizontal Velocity.
Group

I015/601/COGHHV/X

description: Correlation of Geometric Height converted into metres and Horizontal Velocity of (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/601/COGHHV/Y

description: Correlation of Geometric Height converted into metres and Horizontal Velocity of (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/601/COGHHA - Correlation of Geometric Height and Horizontal Acceleration

description: Correlation of Geometric Height converted into metres and Horizontal Acceleration of X/Y-components.
Group

I015/601/COGHHA/X

description: Correlation of Geometric Height converted into metres and Horizontal Acceleration of (X-component), in Two's complement

Element
bit size: 8

Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/601/COGHHA/Y

description: Correlation of Geometric Height converted into metres and Horizontal Acceleration of (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/602 - Horizontal Velocity Information

definition: Magnitude of the Horizontal Velocity Vector.

Compound

I015/602/HV - Horizontal Velocity Vector

description: Horizontal velocity vector expressed in target centric Cartesian coordinates.
Group

I015/602/HV/X

description: Horizontal Velocity (X-component), in Two's complement

Element
bit size: 20
Signed quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
>= -5242.87
<= 5242.87

I015/602/HV/Y

description: Horizontal Velocity in (Y-component), in Two's complement

Element
bit size: 20
Signed quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
>= -5242.87
<= 5242.87

I015/602/RSHV - Horizontal Velocity Resolution

description: Minimum difference in Horizontal Velocity at which a sensor system is able to distinguish two targets with otherwise identical parameters in range and angular domain (under ideal measurement).
Group

I015/602/RSHV/X

description: Horizontal velocity resolution of the target in target centric Cartesian coordinates (X-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
<= 655.35

I015/602/RSHV/Y

description: Horizontal velocity resolution of the target in target centric Cartesian coordinates (Y-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: ”m/s”
<= 655.35

I015/602/RSHV/CORSHVXY

description: Correlation of horizontal position resolution of X and Y components, in Two’s complement.

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

Note: The velocity resolution describes the capability of a sensor to be able to separate two closely located targets in the velocity domain. The velocity resolution for an INCS system depends on the signal integration time as well as the transmit frequency. A longer integration time or a higher transmit frequency leads to a better velocity resolution. Depending on the specific INCS application, a better velocity resolution may be more important than a high sensor refresh rate. The velocity resolution might also alleviate the limitations on the range resolution, in order to make small bandwidth applications possible.

I015/602/SDHV - Horizontal Velocity Precision

description: Root-mean-square (rms) error of the Horizontal Velocity estimate provided by a sensor system.
Group

I015/602/SDHV/X

description: Standard Deviation of horizontal velocity (X-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: ”m/s”
<= 655.35

I015/602/SDHV/Y

description: Standard Deviation of horizontal velocity (Y-component)

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: ”m/s”
<= 655.35

I015/602/SDHV/COHVXY

description: Correlation of standard deviation of horizontal velocity of X and Y components, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/602/COHVHP - Correlation of Horizontal Velocity and Horizontal Position

description: Correlation of the errors associated with the estimates of Horizontal Velocity and Horizontal Position provided by a sensor system.
Group

I015/602/COHVHP/COHVXHPX

description: Correlation of Horizontal Velocity (X-component) and Horizontal Position (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/602/COHVHP/COHVXHPY

description: Correlation of Horizontal Velocity (X-component) and Horizontal Position (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/602/COHVHP/COHVYHPX

description: Correlation of Horizontal Velocity (Y-component) and Horizontal Position (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/602/COHVHP/COHVYHPY

description: Correlation of Horizontal Velocity (Y-component) and Horizontal Position (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/603 - Horizontal Acceleration Information

definition: Magnitude of the Horizontal Acceleration Vector.

Compound

I015/603/HA - Horizontal Acceleration Vector

description: Horizontal acceleration vector expressed in target centric local Cartesian coordinates, in Two's complement representation.
Group

I015/603/HA/X

description: Horizontal Acceleration (X-component), in Two's complement

Element
bit size: 12
Signed quantity
LSB = $1/2^4$ m/s² ≈ $6.25e-2$ m/s²
unit: ”m/s²”
>= -128.0
<= 128.0

I015/603/HA/Y

description: Horizontal Acceleration (Y-component), in Two’s complement

Element
bit size: 12
Signed quantity
LSB = $1/2^4$ m/s² ≈ $6.25e-2$ m/s²
unit: ”m/s²”
>= -128.0
<= 128.0

I015/603/SDHA - Horizontal Acceleration Precision

description: Root-mean-square (rms) error of the Horizontal Acceleration estimate provided by a sensor system.

Group

I015/603/SDHA/X

description: Standard Deviation of Horizontal Acceleration (X-component)

Element
bit size: 12
Unsigned quantity
LSB = $1/2^4$ m/s² ≈ $6.25e-2$ m/s²
unit: ”m/s²”
<= 255.9375

I015/603/SDHA/Y

description: Standard Deviation of Horizontal Acceleration (Y-component)

Element
bit size: 12
Unsigned quantity
LSB = $1/2^4$ m/s² ≈ $6.25e-2$ m/s²
unit: ”m/s²”
<= 255.9375

I015/603/SDHA/COHAXY

description: Correlation of standard deviation of Horizontal Acceleration of X and Y components, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHP - Correlation of Horizontal Acceleration and Horizontal Position

description: Correlation of the errors associated with the estimates of Horizontal Acceleration and Horizontal Position provided by a sensor system.

Group

I015/603/COHAHP/COHAXHPX

description: Correlation of Horizontal Acceleration (X-component) and Horizontal Position (X-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHP/COHAXHPY

description: Correlation of Horizontal Acceleration (X-component) and Horizontal Position (Y-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHP/COHAYHPX

description: Correlation of Horizontal Acceleration (Y-component) and Horizontal Position (X-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHP/COAYHPY

description: Correlation of Horizontal Acceleration (Y-component) and Horizontal Position (Y-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHV - Correlation of Horizontal Acceleration and Horizontal Velocity

description: Correlation of the errors associated with the estimates of Horizontal Acceleration and Horizontal Velocity provided by a sensor system.
Group

I015/603/COHAHV/COHAXHVX

description: Correlation of Horizontal Acceleration (X-component) and Horizontal Velocity (X-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHV/COHAXHVY

description: Correlation of Horizontal Acceleration (X-component) and Horizontal Velocity (Y-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/603/COHAHV/COHAYHVX

description: Correlation of Horizontal Acceleration (Y-component) and Horizontal Velocity (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/603/COHAHV/COHAYHVY

description: Correlation of Horizontal Acceleration (Y-component) and Horizontal Velocity (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604 - Vertical Velocity Information

definition: Vertical velocity as given by the rate of change of the Geometric Height.

Compound

I015/604/VV - Vertical Velocity

description: Vertical velocity as given by the rate of change of the Geometric Height expressed in Two's Complement.

Element
bit size: 24
Signed quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
>= -83886.07
<= 83886.07

Note: Positive values indicates climbing target and negative values indicates descending target.

I015/604/RSVV - Vertical Velocity Resolution

description: Minimum difference in Vertical Velocity at which a sensor system is able to distinguish two targets with otherwise identical parameters in range and angular domain (under ideal measurement).

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
<= 655.35

I015/604/SDVV - Vertical Velocity Precision

description: Root-mean-square (rms) error of the Vertical Velocity estimate provided by a sensor system.

Group

I015/604/SDVV/SDVV

description: Standard Deviation of Vertical Velocity

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: "m/s"
<= 655.35

I015/604/SDVV/COVVGH

description: Correlation of Vertical Velocity and Geometric Height, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHP - Correlation of Vertical Velocity and Horizontal Position

description: Correlation of the errors associated with the estimates of Vertical Velocity converted in to metres/sec and Horizontal Position provided by a sensor system.
Group

I015/604/COVVHP/X

description: Correlation of Vertical Velocity and Horizontal Position (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHP/Y

description: Correlation of Vertical Velocity and Horizontal Position (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHV - Correlation of Vertical Velocity and Horizontal Velocity

description: Correlation of the errors associated with the estimates of Vertical Velocity converted in to metres/sec and Horizontal Velocity provided by a sensor system.
Group

I015/604/COVVHV/X

description: Correlation of Vertical Velocity and Horizontal Velocity (X-component)), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHV/Y

description: Correlation of Vertical Velocity and Horizontal Velocity (Y-component)), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHA - Correlation of Vertical Velocity and Horizontal Acceleration

description: Correlation of the errors associated with the estimates of Vertical Velocity converted in to metres/sec and Horizontal Acceleration provided by a sensor system.
Group

I015/604/COVVHA/X

description: Correlation of Vertical and Horizontal Acceleration (X-component)), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/604/COVVHA/Y

description: Correlation of Vertical Velocity and Horizontal Acceleration (Y-component)), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/605 - Vertical Velocity Information

definition: Compound data item, comprising a primary subfield of one-octet, followed by one or more defined subfields.

Compound

I015/605/VA - Vertical Acceleration

description: Vertical acceleration information expressed in Two's complement.

Element
bit size: 16
Signed quantity
LSB = $1/100$ m/s² $\approx 1.0e-2$ m/s²
unit: "m/s²"
>= -327.67
<= 327.67

Note: Positive values indicates accelerating during climb or descent and negative values indicates deceleration during climb or descent.

I015/605/RSVA - Vertical Acceleration Precision

description: Root-mean-square (rms) error of the Vertical Acceleration estimate provided by a sensor system.
Group

I015/605/RSVA/SDVA

description: Standard Deviation of Vertical Acceleration

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ m/s² $\approx 1.0e-2$ m/s²
unit: "m/s²"
<= 167772.15

I015/605/RSVA/COVAGH

description: Correlation of Vertical Acceleration and Geometric Height, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/RSVA/COVAVV

description: Correlation of Vertical Acceleration and Vertical Velocity, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/COVAHP - Correlation of Vertical Acceleration and Horizontal Position

description: Correlation of the errors associated with the estimates of Vertical Acceleration and Horizontal Position provided by a sensor system.

Group

I015/605/COVAHP/X

description: Correlation of Vertical Acceleration and Horizontal Position (X-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/COVAHP/Y

description: Correlation of Vertical Acceleration and Horizontal Position (Y-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/COVAHV - Correlation of Vertical Acceleration and Horizontal Velocity

description: Correlation of the errors associated with the estimates of Vertical Acceleration and Horizontal Velocity provided by a sensor system.

Group

I015/605/COVAHV/X

description: Correlation of Vertical Acceleration and Horizontal Velocity (X-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/COVAHV/Y

description: Correlation of Vertical Acceleration and Horizontal Velocity (Y-component), in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/605/COVAHA - Correlation of Vertical Acceleration and Horizontal Acceleration

description: Correlation of the errors associated with the estimates of Vertical Acceleration and Horizontal Acceleration provided by a sensor system.
Group

I015/605/COVAHA/X

description: Correlation of Vertical Acceleration and Horizontal Acceleration (X-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/605/COVAHA/Y

description: Correlation of Vertical Acceleration and Horizontal Acceleration (Y-component), in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/625 - Range Information

definition: The targets range information is given relative to the sensor reference point(s).

Compound

I015/625/R - Range

description: Measured range between a target object and a pre-defined point associated with the sensor system (e.g., for a monostatic radar system the phase centre of the antenna aperture) or measured bistatic range between a pre-defined point associated with the transmitter station, the target object position and a pre-defined point associated with the sensor system (e.g., for a bistatic radar system the phase centres of the transmitter and receiver antenna aperture).

Element
bit size: 24
Signed quantity
LSB = $1/10$ m ≈ 0.1 m
unit: "m"
>= -838860.7
<= 838860.7

I015/625/RSR - Range Resolution

description: Minimum difference in Range at which a sensor system is able to distinguish two targets with otherwise identical parameters in velocity and angular domain (under ideal measurement).

Element
bit size: 24
Unsigned quantity
LSB = $1/10$ m ≈ 0.1 m
unit: "m"
<= 1677721.5

Note: This may differ from the cell size applied within the Sensor.

I015/625/SDR - Range Precision

description: Root-mean-square (rms) error of the Range estimate provided by a sensor system.

Element
bit size: 24
Unsigned quantity
LSB = $1/10$ m ≈ 0.1 m
unit: ”m”
<= 1677721.5

I015/625/RR - Range Rate

description: The range rate is derived from different range measurements.

Element
bit size: 24
Signed quantity
LSB = $1/10$ m/s ≈ 0.1 m/s
unit: ”m/s”
>= -838860.7
<= 838860.7

Note: The range rate is the first derivative computed from the range. In contrast, the Doppler velocity in I015/626 is actually measured through Doppler.

I015/625/RSRR - Range Rate Resolution

description: Minimum difference in Range Rate at which a sensor system is able to distinguish two targets with otherwise identical parameters in position and angular domain (under ideal measurement).

Element
bit size: 24
Unsigned quantity
LSB = $1/10$ m/s ≈ 0.1 m/s
unit: ”m/s”
<= 1677721.5

I015/625/SDRR - Range Rate Precision

description: Root-mean-square (rms) error of the Range Rate estimate provided by a sensor system.
Group

I015/625/SDRR/SDRR

description: Standard Deviation of Range Rate

Element
bit size: 24
Unsigned quantity
LSB = $1/10$ m/s ≈ 0.1 m/s
unit: ”m/s”
<= 1677721.5

I015/625/SDRR/CORRR

description: Correlation of Range Rate and Range, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7$ ≈ $7.8125e-3$
unit: ””

I015/625/RA - Range Acceleration

description: The range acceleration is derived from different range rates.

Element
bit size: 16
Signed quantity
LSB = $1/2^6$ m/s² ≈ $1.5625e-2$ m/s²
unit: "m/s²"
>= -512.0
<= 512.0

I015/625/SDRA - Range Acceleration Precision

description: Root-mean-square (rms) error of the Range Acceleration determined by the sensor system.
Group

I015/625/SDRA/SDRA

description: Standard Deviation of Range Acceleration

Element
bit size: 16
Unsigned quantity
LSB = $1/2^7$ m/s² ≈ $7.8125e-3$ m/s²
unit: "m/s²"
<= 512.0

I015/625/SDRA/CORAR

description: Correlation of Range Acceleration and Range, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7$ ≈ $7.8125e-3$
unit: ""

I015/625/SDRA/CORARR

description: Correlation of Range Acceleration and Range Rate, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7$ ≈ $7.8125e-3$
unit: ""

Note 1: Depending upon its design the manner in which the positional data is declared by an INCS sensor may be expressed in WGS-84 (I015/ 600) or sensor centric coordinate system based upon the 'system reference point' of the sensor (I015/625).

Note 2: The optional 'precision' fields (in Data Items I015/600 – I015/625) provide a measure of the accuracy the INCS system has assigned to positional data contained in the target report. Such information can be used to improve the quality with which the INCS target report data is integrated in to the subsequent processing stages of the ATM infrastructure. However a consideration of the sensor characteristics and capabilities, the manner in which INCS data is used operationally, the weighting assigned to INCS data within a multi-sensor tracker and the credibility assigned to the covariance data items should be made to support the decision of whether these optional covariance data items are required or whether the basic data items provide sufficient performance. Whilst not an operational consideration the additional costs that may be associated with the provision and use of such data items should also be weighed against the potential performance benefits that would be achieved through the inclusion of these Data Items in performance specifications.

Note 3: If I015/020 MoMu indicates that the target report is Bi-Static (MoMu =1) then the range information is the difference between the path from the transmitter to target to the receiver less the distance between the transmitter and receiver. In this case, the reference points referred to above are the positions of the transmitter and receiver. If I015/020 MoMu indicates that the target report is Mono-Static (MoMu =0)

then the range information is the distance between the sensor and the target. In this case, the reference point referred to above is the position of the mono-static sensor.

Note 4: The meaning of range in Category 015 is significantly broader than the traditional hence the different INCS working principles. This is especially true for bi-static and multi-static radars. For readability the field is still called range and not mono-/bi-/multi-static range or pseudo-range. For radar the measured range is calculated from time differences of signals assumed to be transmitted/received at the sensor reference point(s). As noted above for bi-static radars the reference points are the positions of the transmitter and receiver. Moreover multi-static radars may receive identical signals from other transmitters than the assumed reference transmitter (e.g. passive INCS which are using single frequency networks as illuminators) and therefore may calculate negative values for bi-static range measurement data.

I015/626 - Doppler Information

definition: Doppler measurement of the Target.

Compound

I015/626/DV - Doppler Velocity

description: Radial velocity or bistatic velocity of a target object measured by a sensor system via a corresponding Doppler frequency shift.

Element
bit size: 24
Signed quantity
LSB = $1/100$ m/s $\approx 1.0e-2$ m/s
unit: ”m/s”
>= -83886.07
<= 83886.07

Note: The radial velocity is the magnitude of the 3-dimensional velocity vector (i.e., the time derivative of the 3-dimensional position vector) projected onto the line between target object and sensor. The bistatic velocity is the magnitude of the 3-dimensional velocity vector projected onto the line between transmitter station and target object plus the magnitude of the 3-dimensional velocity vector projected onto the line between target object and sensor.

I015/626/SDDV - Precision of Doppler Velocity

description: Root-mean-square (rms) error of the Doppler Velocity measured by the sensor system.

Element
bit size: 16
Unsigned quantity
LSB = $1/2^6$ m/s $\approx 1.5625e-2$ m/s
unit: ”m/s”
<= 1024.0

I015/626/DA - Doppler Acceleration

description: Radial acceleration or bistatic acceleration of a target object measured by a sensor system via a corresponding Doppler frequency shift and a subsequent difference operation.

Element
bit size: 16
Signed quantity
LSB = $1/2^6$ m/s² $\approx 1.5625e-2$ m/s²
unit: ”m/s²”
>= -512.0
<= 512.0

Note: The radial acceleration is the magnitude of the 3-dimensional acceleration vector (i.e., the time derivative of the 3-dimensional velocity vector) projected onto the line between target object and sensor. The bistatic acceleration is the magnitude of the 3-dimensional acceleration vector projected onto the line between transmitter station and target object plus the magnitude of the 3-dimensional acceleration vector projected onto the line between target object and sensor.

I015/626/SDDA - Precision of Doppler Acceleration

description: Root-mean-square (rms) error of the Doppler Velocity measured by the sensor system.
Group

I015/626/SDDA/SDDA

description: Standard Deviation of Doppler Acceleration

Element
bit size: 16
Unsigned quantity
LSB = $1/2^6$ m/s² ≈ $1.5625e-2$ m/s²
unit: "m/s²"
<= 1024.0

I015/626/SDDA/CODADV

description: Correlation of Doppler Acceleration and Doppler Velocity, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/626/CODVR - Correlation of Doppler Velocity and Range

description: Correlation of Doppler Velocity and Range (e.g. bistatic range).

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/626/CODVRR - Correlation of Doppler Velocity and Range Rate

description: Correlation of Doppler Velocity and Range Rate.

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/626/CODVRA - Correlation of Doppler Velocity and Range Acceleration

description: Correlation of Doppler Velocity and Range (e.g. bistatic range).

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/626/CODAR - Correlation of Doppler Acceleration and Range

description: Correlation of Doppler Acceleration and Range (e.g. bistatic range).

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/626/CODARR - Correlation of Doppler Acceleration and Range Rate

description: Correlation of Doppler Acceleration and Range Rate.

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/626/CODARA - Correlation of Doppler Acceleration and Range Acceleration

description: Correlation of Doppler Acceleration and Range Acceleration.

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ””

I015/627 - Azimuth Information

definition: Azimuth information that is provided relative to the sensor or component reference point.

Compound

I015/627/AZ - Azimuth

description: Target angle relative to geographic North in the local reference system centred on the sensor.

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° $\approx 5.4931640625e-3$ °
unit: ”°”
>= 360.0

Note: The azimuth shall increment in a clockwise manner relative to geographic North.

I015/627/RSAZ - Azimuth Resolution

description: Minimum angle in order to separate targets by the sensor in the azimuth dimension.

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ ° $\approx 6.866455078125e-4$ °
unit: ”°”
<= 45.0

I015/627/SDASZ - Standard Deviation of Azimuth

description: Estimated standard deviation of the azimuth angle.

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ ° $\approx 6.866455078125e-4$ °
unit: ”°”
<= 45.0

I015/627/AZR - Azimuth Rate

description: Rate of change of the azimuth angle.

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ ° ≈ $2.74658203125e-3$ °
unit: ”°”
>= -90.0
<= 90.0

I015/627/SDAZR - Standard Deviation of Azimuth Rate

description: Estimated standard deviation of the azimuth angle rate.
Group

I015/627/SDAZR/SDAZR

description: Standard Deviation of Azimuth Rate

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ ° ≈ $6.866455078125e-4$ °
unit: ”°”
<= 45.0

I015/627/SDAZR/COAZRAZ

description: Correlation of Azimuth Rate and Azimuth, in Two’s complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7$ ≈ $7.8125e-3$
unit: ””

I015/627/AZEX - Azimuth Extent

description: Target size in the azimuth angle dimension. The target extends between start angle and end angle traversed clockwise.
Group

I015/627/AZEX/S

description: Azimuth Extent Start

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: ”°”
<= 360.0

I015/627/AZEX/E

description: Azimuth Extent End

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: ”°”
<= 360.0

Note: The Sensor Reference Point is detailed in ASTERIX Category 016 – where there is also provision for including the reference points for the transmitter(s) and receiver(s) that are used within the sensor configuration. The Sensor Reference Point is also contained in ASTERIX Category 025.

I015/628 - Elevation Information

definition: Information related to the elevation angle provided by the sensor. (Predominantly used by electro-optic sensors).

Compound

I015/628/EL - Elevation

description: The elevation shall be given with respect to the horizontal plane of the sensor expressed in Two's Complement.

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ ° ≈ $2.74658203125e-3$ °
unit: "°"
>= -90.0
<= 90.0

Note: The elevation shall be given with respect to the local WGS-84 tangential plane of the receiver dedicated by I015/400. Note: For targets above the horizontal plane the elevation angle is positive and for targets below negative.

I015/628/RSEL - Elevation Resolution

description: Minimum angle in order to separate targets by the sensor in the elevation dimension.

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ ° ≈ $6.866455078125e-4$ °
unit: "°"
<= 45.0

I015/628/SDEL - Standard Deviation of Elevation

description: Estimated standard deviation of the elevation angle.

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ ° ≈ $6.866455078125e-4$ °
unit: "°"
<= 45.0

I015/628/ER - Elevation Rate

description: Rate of change of the elevation angle.

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ °/s ≈ $2.74658203125e-3$ °/s
unit: "°/s"
>= -90.0
<= 90.0

I015/628/SDER - Standard Deviation of Elevation Rate

description: Estimated standard deviation of the elevation angle rate.
Group

I015/628/SDER/SDELR

description: Standard Deviation of Elevation Rate

Element
bit size: 16
Unsigned quantity
LSB = $45/2^16$ °/s ≈ $6.866455078125e-4$ °/s
unit: "°/s"

I015/628/SDER/COELREL

description: Correlation of Elevation Rate and Elevation, in Two's complement

Element
bit size: 8
Signed quantity
LSB = $1/2^7 \approx 7.8125e-3$
unit: ""

I015/628/ELEX - Elevation Extent

description: Target size in the elevation angle dimension. The target extends between start angle and end angle.
Group

I015/628/ELEX/S

description: Elevation Extent Start, in Two's complement

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ ° $\approx 2.74658203125e-3$ °
unit: "°"
>= -90.0
<= 90.0

I015/628/ELEX/E

description: Elevation Extent End, in Two's complement

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ ° $\approx 2.74658203125e-3$ °
unit: "°"
>= -90.0
<= 90.0

I015/630 - Path Quality

definition: Measure characterising the signal quality associated with a specific target echo signal.

Compound

I015/630/DPP - Direct Path - Power

description: Signal power measured for the direct signal received from a specific transmitter station.

Element
bit size: 8
Signed quantity
LSB = 1 dB ≈ 1.0 dB
unit: "dB"
>= -128.0
<= 127.0

I015/630/DPS - Direct Path - Signal to Noise Ratio (SNR)

description: Signal to noise ratio measured for the direct signal received from a specific transmitter station.

Element
bit size: 8
Signed quantity
LSB = 1 dB ≈ 1.0 dB
unit: "dB"
>= -128.0
<= 127.0

I015/630/RPP - Reflected Path - Power

description: Signal power measured for a specific target echo signal found within range-Doppler matrix (associated with a specific transmitter station).

Group

Spare bits: 7

I015/630/RPP/RPP

description: Power of reflected path, in Two's complement"

Element
bit size: 9
Signed quantity
LSB = 1 dB ≈ 1.0 dB
unit: "dB"
>= -256.0
<= 255.0

I015/630/RPS - Reflected Path - Signal to Noise Ratio (SNR)

description: Signal to noise ratio measured for a specific target echo signal found within range-Doppler matrix (associated with a specific transmitter station).

Element
bit size: 8
Signed quantity
LSB = 1 dB ≈ 1.0 dB
unit: "dB"
>= -128.0
<= 127.0

Notes: Some INCS sensors may be capable of outputting an indication of the signal quality based upon the received echo signal strength for that target. Before including the provision of such data items in the technical specification, it is advised that the cost and operational benefits of the availability of such data is assessed.

I015/631 - Contour (Azimuth, Elevation Angle, Range Extent)

definition: Azimuth, elevation angles and range extent of all elementary presences constituting a plot.

Repetitive

Regular, 1 byte(s) REP field size.

Group

I015/631/AZCON

description: Azimuth Contour

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: "°"
>= 360.0

I015/631/ELCON

description: Elevation Contour, in Two's complement

Element
bit size: 16
Signed quantity
LSB = $180/2^16$ ° ≈ $2.74658203125e-3$ °
unit: "°"
<= -90.0
>= 90.0

I015/631/RGCONSTOP

description: Range Contour Stop

Element
bit size: 16
Unsigned quantity
LSB = $10000/2^16$ m ≈ 0.152587890625 m
unit: ”m”

I015/631/RGCONSTART

description: Range Contour Start

Element
bit size: 16
Unsigned quantity
LSB = $10000/2^16$ m ≈ 0.152587890625 m
unit: ”m”

Note 1: The azimuth shall increment in a clockwise manner relative to geographic North centred at the System Reference Point.

Note 2: The elevation shall be given with respect to the local WGS-84 tangential plane of the receiver dedicated by I015/400.

Note 3: If populated, the range contour requires a start and stop point. The stop point is to be greater or equal than the start point.

I015/SP - Special Purpose Field

definition: Special Purpose Field

Explicit (SpecialPurpose)

User Application Profile

- 1: `I015/010` - Data Source Identifier
- 2: `I015/000` - Message Type
- 3: `I015/015` - Service Identification
- 4: `I015/020` - Target Report Descriptor
- 5: `I015/030` - Warning/Error Conditions
- 6: `I015/145` - Time of Applicability
- 7: `I015/161` - Track/Plot Number
- (FX) - Field extension indicator
- 8: `I015/170` - Track/Plot Status
- 9: `I015/050` - Update Period
- 10: `I015/270` - Target Size & Orientation
- 11: `I015/300` - Object Classification
- 12: `I015/400` - Measurement Identifier
- 13: `I015/600` - Horizontal Position Information
- 14: `I015/601` - Geometric Height Information
- (FX) - Field extension indicator
- 15: `I015/602` - Horizontal Velocity Information
- 16: `I015/603` - Horizontal Acceleration Information
- 17: `I015/604` - Vertical Velocity Information
- 18: `I015/605` - Vertical Velocity Information
- 19: `I015/480` - Associations
- 20: `I015/625` - Range Information
- 21: `I015/626` - Doppler Information
- (FX) - Field extension indicator
- 22: `I015/627` - Azimuth Information
- 23: `I015/628` - Elevation Information
- 24: `I015/630` - Path Quality
- 25: `I015/631` - Contour (Azimuth, Elevation Angle, Range Extent)

- 26: `I015/SP` - Special Purpose Field
- *Spare*
- *Spare*
- (FX) - Field extension indicator